

Table of Contents

1. Game Overview	3
1.1 Design Pillars	3
1.2 Target Audience	3
2. Core Game Loop	4
3. Gacha Pull System	4
3.1 Banner Types	4
3.2 Pull Rates	5
3.3 Pull Costs	5
3.4 Pity System	5
3.5 Duplicate Handling	6
4. Studio Room Management	6
4.1 Station Types	6
4.2 Station Upgrades	6
4.3 Resource Output Calculation	7
5. Currency System	7
5.1 Starting Resources	8
5.2 Daily Login Bonus	8
6. Character Progression	8
6.1 Character Levelling	8
6.2 Character Ascension	9
7. Studio Level and Feature Unlocks	9
7.1 Future Feature: Odekake (Going Out)	10
8. Collection Gallery	10
8.1 Gallery Features	10
9. UI/UX Design	11

9.1 Visual Theme	11
9.2 Navigation	11
9.3 Card Design	12
9.4 Responsive Design	12
10. Technical Architecture	12
10.1 Technology Stack	12
10.2 File Structure	13
10.3 Save/Load System	14
10.4 Game Tick System	14
10.5 Hybrid Save Sync System	14
11. Character Data	16
11.1 Data Schema	16
11.2 Agency Distribution	16
12. Sprint Development Plan	17
12.1 Development Workflow	18
13. Appendix: Key Data Structures	18
13.1 Game State Object	18

1. Game Overview

MyVT Gacha Collection is a browser-based idle gacha collection game themed around the Malaysian VTuber (MyVT) community. Players collect virtual cards featuring real Malaysian VTubers by pulling from gacha banners, then manage a virtual studio where assigned VTubers generate resources passively. The game combines the excitement of gacha pulls with idle management mechanics, creating an engaging gameplay loop that rewards both short sessions and long-term dedication.

The game is built entirely with vanilla HTML, CSS, and JavaScript, making it lightweight, accessible, and hostable on any static web hosting platform such as GitHub Pages or Neocities. No backend server or database is required; all game state is persisted using the browser's localStorage. The roster of characters is sourced from hololist.net, which currently lists 319 Malaysian VTubers spanning independent creators and members of various agencies including VGakuenLive, HoloDream, MyHolo TV (VILIT), Projek Hikayat, and Phase Connect.

The visual style follows an anime-game aesthetic with a dark theme using purple and blue accent colours. Character cards feature the VTubers' portrait images sourced from hololist, with decorative frames that change based on the card's variant tier. The UI is designed to be responsive, working seamlessly on both desktop browsers and mobile devices, ensuring that members of the MyVT community can enjoy the game regardless of their preferred platform.

1.1 Design Pillars

- **Accessibility:** No install required, runs in any modern browser, zero cost to play.
- **Community Celebration:** Showcases Malaysian VTubers and raises awareness of the scene.
- **Idle-Friendly:** Progress continues even when offline, rewarding regular check-ins.
- **Collection Satisfaction:** 319 characters with variant upgrades provide long-term goals.
- **Skill Development:** Serves as a learning project for web game development.

1.2 Target Audience

The primary audience consists of members of the MyVT community, including VTubers themselves, their fans, and supporters of the Malaysian VTuber scene. Secondary audience members include general gacha game enthusiasts and idle game players who may discover the game through community sharing. The game is designed to be approachable for newcomers while offering enough depth to retain experienced gacha players. All text in the game is presented in English, reflecting the

multilingual nature of the Malaysian VTuber community.

2. Core Game Loop

The core gameplay loop of MyVT Gacha Collection revolves around a cycle of pulling characters, assigning them to studio stations, earning resources passively, and reinvesting those resources to pull more characters and strengthen existing ones. This loop is designed to be simple enough to understand in seconds but deep enough to sustain engagement over weeks and months of play.

Step	Action	Description
1	Pull	Spend Stars on gacha banners to acquire VTuber character cards.
2	Assign	Place collected characters into Studio Room stations to generate resources.
3	Earn	Resources accumulate passively (idle), even while the player is offline.
4	Level Up	Spend Star Dust and Stars to increase character levels, boosting station output.
5	Ascend	Spend Star Fragments to upgrade Normal cards to SR and SR to SSR variants.
6	Expand	Upgrade Studio Level to unlock new stations, slots, and future features.
7	Repeat	Continue pulling to fill collection gaps; aim for 100% SSR completion.

Table 1: The Seven-Step Core Game Loop

New players begin with 1,000 Stars, enough for approximately 10 single pulls on the standard banner. This immediate access to gacha pulls is intentional, as it allows players to experience the core excitement of the game within seconds of starting. The first few pulls will typically yield Normal variant characters, which the player can immediately assign to the Stream Room station to begin generating Stars passively. This establishes the resource generation foundation that fuels the rest of the gameplay loop.

3. Gacha Pull System

3.1 Banner Types

The gacha system features two types of banners that players can pull from. The Standard Banner contains all 319 Malaysian VTubers from the hololist roster, making it the primary source of new characters. The Featured Banner is a rotating limited-time banner that highlights 3-4 specific VTubers with increased drop rates,

providing focused collection opportunities for players seeking specific characters.

Banner Type	Pool Size	Featured Characters	Rotation
Standard	319	None (equal rates)	Permanent
Featured	319	3-4 rate-up VTubers	Weekly

Table 2: Banner Types Overview

3.2 Pull Rates

Each pull yields one character card with a randomly determined variant tier. The variant system does not affect which character is obtained; rather, it determines the rarity and visual quality of that character's card. This design means that every character is equally obtainable, and the excitement of pulling comes from the variant tier reveal rather than character exclusivity.

Variant	Drop Rate	Card Frame	Pull Guarantee
Normal	82%	Grey border, standard art	Always obtainable
SR	15%	Silver border with glow, alt art	1 per 10x pull
SSR	3%	Gold border with shine, premium art	Pity at 50 pulls

Table 3: Pull Rates by Variant Tier

3.3 Pull Costs

Pull Type	Cost (Stars)	Guarantee
Single Pull (1x)	100	None
Multi Pull (10x)	1,000	At least 1 SR or above

Table 4: Pull Costs

3.4 Pity System

A pity system ensures that players are guaranteed an SSR variant after 50 consecutive pulls without obtaining one. The pity counter tracks pulls across all banners and resets every time an SSR variant is pulled, regardless of which banner it came from. This prevents excessively unlucky streaks and ensures that dedicated players always make meaningful progress toward high-tier characters. Additionally, the 10x multi-pull guarantees at least one SR or above card, providing a reliable floor for multi-pull investments.

3.5 Duplicate Handling

When a player pulls a character they already own (same character and same variant tier), the duplicate is converted into Shards specific to that character. Shards serve as a universal resource that can be exchanged for Star Dust or Star Fragments at a conversion rate displayed in the in-game shop. If a player already owns a character at SSR tier and pulls that same character at any tier again, the duplicate is automatically converted into bonus Stars instead of Shards, providing a meaningful endgame resource sink.

4. Studio Room Management

The Studio Room is the management hub of MyVT Gacha Collection. It is a visual space where players assign their collected VTuber characters to various Stations, each of which produces a different type of resource. The Studio provides the idle management layer that transforms the game from a simple pull simulator into an engaging resource management experience. Players must strategically decide which characters to assign to which stations based on their current needs and long-term goals.

4.1 Station Types

Each station represents a different activity that VTubers might engage in, and each produces a unique resource. Players start with the Stream Room station unlocked and gradually unlock additional stations by increasing their Studio Level. Every station can hold exactly one assigned VTuber at a time, and the player may reassign characters freely with no cooldown or penalty. The output of each station scales with the assigned character's variant tier and level, as well as the station's own upgrade level.

Station	Resource	Primary Use	Unlock
Stream Room	Stars	Gacha pulls, station upgrades	Studio Lv 1
Creative Corner	Star Dust	Character levelling XP	Studio Lv 2
Practice Hall	Star Fragments	Ascension material crafting	Studio Lv 4
Lounge	Bond Points	Studio Level EXP	Studio Lv 6

Table 5: Station Types and Their Resources

4.2 Station Upgrades

Each station can be upgraded from Level 1 through Level 5. Upgrading a station increases its output multiplier, meaning that the same assigned character will

produce more resources per minute at higher station levels. The upgrade costs scale with each level, requiring increasing amounts of Stars and Star Fragments. The output multiplier progresses as follows: Level 1 provides a x1 multiplier, Level 2 provides x1.5, Level 3 provides x2, Level 4 provides x3, and Level 5 provides x5. This exponential scaling means that maxing out a station is a significant long-term investment that dramatically increases resource generation.

Level	Multiplier	Upgrade Cost (Stars)	Upgrade Cost (Fragments)
1	x1	(Base)	(Base)
2	x1.5	500	50
3	x2	2,000	200
4	x3	8,000	800
5	x5	30,000	3,000

Table 6: Station Upgrade Costs and Multipliers

4.3 Resource Output Calculation

The resource output per minute for any station is determined by the following formula: $\text{Output} = \text{Base Rate} \times \text{Character Variant Multiplier} \times \text{Character Level Bonus} \times \text{Station Level Multiplier}$. The base rate differs per station type. Character variant multipliers are x1 for Normal, x2 for SR, and x5 for SSR. The character level bonus is calculated as $1 + (\text{Level} \times 0.02)$, meaning a Level 50 character provides a x2 bonus on top of the variant multiplier. This formula ensures that every aspect of character progression meaningfully contributes to resource generation, making both pulling and upgrading feel rewarding.

5. Currency System

MyVT Gacha Collection features four distinct currencies, each produced by a different studio station and serving a unique purpose within the game economy. This multi-currency design ensures that players must engage with all aspects of the game to progress, preventing the common gacha game problem of a single currency becoming irrelevant after the early game. Each currency feeds into different progression systems, creating interdependencies that make the gameplay loop feel cohesive and rewarding.

Currency	Sy mb ol	Source	Primary Use
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Stars	*	Stream Room, Daily Login, Duplicate SSR	Gacha pulls, Station upgrades
Star Dust	+	Creative Corner, Shard conversion	Character levelling
Star Fragments	#	Practice Hall, Shard conversion	Ascension crafting
Bond Points	~	Lounge station	Studio Level EXP

Table 7: Currency Types, Sources, and Uses

5.1 Starting Resources

New players receive a starting package of 1,000 Stars to immediately begin pulling on banners. No other currencies are provided at the start; players must progress through the early game to unlock stations and begin generating secondary currencies. This design intentionally front-loads the gacha experience, allowing new players to experience the excitement of pulling within their first minute of gameplay. The 1,000 Star starting bonus provides enough for 10 single pulls or 1 multi-pull, giving players a meaningful initial collection to work with.

5.2 Daily Login Bonus

Players receive a Daily Login Bonus of Stars that scales with their consecutive login streak. The bonus starts at 100 Stars for the first day and increases by 50 Stars per consecutive day, capping at 500 Stars per day after a 9-day streak. Missing a day resets the streak to day 1. This system encourages regular engagement without being overly punitive, as even a single day's bonus provides enough for one pull after a few days of accumulation.

6. Character Progression

Characters in MyVT Gacha Collection progress through two interconnected systems: levelling and ascension. Levelling strengthens a character within its current variant tier, increasing its resource output when assigned to a station. Ascension upgrades a character to a higher variant tier (Normal to SR, or SR to SSR), unlocking a new card frame, new artwork, higher level caps, and significantly increased output multipliers. Both systems require different currencies, ensuring that players must engage with multiple station types to fully develop their roster.

6.1 Character Levelling

Each character can be levelled from Level 1 up to a maximum level determined by their current variant tier. Normal variant characters have a level cap of 20, SR

characters can reach Level 35, and SSR characters can reach Level 50. Levelling requires spending Star Dust and Stars, with costs scaling based on the target level. Each level increases the character's resource output by approximately 2% (the level bonus formula is $1 + \text{Level} \times 0.02$), making levelling a consistent and predictable investment.

Variant	Level Cap	Cost per Level (Star Dust)	Cost per Level (Stars)
Normal	20	$10 + (\text{Lvl} \times 5)$	$20 + (\text{Lvl} \times 10)$
SR	35	$30 + (\text{Lvl} \times 10)$	$50 + (\text{Lvl} \times 20)$
SSR	50	$80 + (\text{Lvl} \times 20)$	$100 + (\text{Lvl} \times 40)$

Table 8: Levelling Costs by Variant Tier

6.2 Character Ascension

Ascension is the process of upgrading a character to a higher variant tier. This is the primary long-term progression goal for each character in the collection. To ascend a Normal character to SR, the character must be at maximum level (20) and the player must spend Star Fragments. To ascend an SR character to SSR, the character must be at its maximum SR level (35) and the player must spend a larger quantity of Star Fragments. Ascension changes the character's card frame, unlocks new artwork, increases the variant output multiplier, raises the level cap, and increases Studio EXP contribution.

Ascension	Required Level	Cost (Star Fragments)	New Level Cap
Normal to SR	Lv 20	100	35
SR to SSR	Lv 35	500	50

Table 9: Ascension Requirements

7. Studio Level and Feature Unlocks

The Studio Level serves as the overarching progression gate for the game. Unlike character levels or station upgrades, the Studio Level does not directly improve resource generation rates. Instead, it unlocks new features, stations, and station slots, providing a structured sense of progression that guides players through the game's systems at a measured pace. Studio EXP is earned passively from assigned characters in stations, with SSR characters contributing more EXP than SR, and SR contributing more than Normal. The Studio Level ensures that new players are not overwhelmed by too many systems at once while providing experienced players with

clear goals to work toward.

Studio Lv	Unlocks	EXP Required
1	Stream Room + 2 station slots (starting state)	0
2	Creative Corner station unlocked	200
3	3rd station slot	600
4	Practice Hall station unlocked	1,500
5	4th station slot	3,500
6	Lounge station unlocked	7,000
7	Character detail stats in gallery	12,000
8	5th station slot	20,000
9	Collection filter by agency	35,000
10	Odekake (Going Out) feature unlocked	60,000

Table 10: Studio Level Progression and Unlocks

7.1 Future Feature: Odekake (Going Out)

At Studio Level 10, players unlock the Odekake feature, inspired by the training camp system in Uma Musume. Odekake allows players to send a character on a timed "outing" lasting between 1 and 4 hours. During this time, the character cannot be assigned to any station. When the outing concludes, the character returns with Bond Points, Studio EXP, and occasionally rare crafting materials. Repeated outings with the same character build a Bond Level, which unlocks character-specific bonuses such as increased output multipliers or reduced ascension costs. This feature adds an emotional attachment layer to the game, encouraging players to invest in their favourite characters beyond mere numerical optimization.

8. Collection Gallery

The Collection Gallery is the player's visual archive of all 319 Malaysian VTubers in the game. It displays every character as a card in a responsive grid layout, with distinct visual indicators showing which characters the player owns, which variant tiers they have obtained, and which characters remain undiscovered. The gallery serves as both a progress tracker and a showcase of the player's collection achievements.

8.1 Gallery Features

- **Grid View:** All 319 characters displayed as cards with portrait images from hololist.net.
- **Ownership Filter:** Filter to show Owned only, Unowned only, or All characters.
- **Agency Filter:** Filter by agency (Independent, VGakuenLive, VILIT, etc.). Unlocked at Studio Level 9.
- **Variant Filter:** Filter by highest owned variant (Normal, SR, SSR).
- **Character Detail Popup:** Click any card to view full details including all owned variants, current level, ascension status, and stat breakdown.
- **Collection Completion:** Percentage tracker showing overall completion and per-variant completion (e.g., 45/319 SSR, 120/319 SR, 280/319 Normal).

9. UI/UX Design

9.1 Visual Theme

The game uses a dark theme with purple and blue accent colours, following the visual language common in anime-style gacha games. The primary background is a deep dark grey (approximately #0e0e0c), with card backgrounds in slightly lighter tones to create visual depth. Purple accents (#896fd9) are used for interactive elements, headers, and SSR indicators, while teal/green accents (#5dd499) highlight positive feedback such as resource gains and collection milestones. Gold tones are reserved for SSR-tier elements to convey rarity and premium quality.

9.2 Navigation

The user interface employs a tab-based navigation system with four primary sections: Home, Pull, Collection, and Studio. A persistent top navigation bar displays these tabs along with the player's current resource counts (Stars, Star Dust, Star Fragments, Bond Points) for quick reference. The navigation is designed to be thumb-friendly on mobile devices, with tabs positioned at the bottom of the screen on narrow viewports and at the top on wider screens. A profile/stats overlay is accessible from a settings icon in the navigation bar.

Tab	Content	Key Interactions
Home	Player stats, daily login, offline earnings collection, studio overview	Claim offline earnings, view progress summary
Pull	Gacha banner selection, pull animation, result display	Single/multi pull, banner switching, pull history

Collecti on	Character gallery grid with filters	Filter, sort, view character details
Studio	Studio Room with station slots and assignment interface	Assign/reassign characters, upgrade stations

Table 11: Navigation Tabs and Their Content

9.3 Card Design

Character cards are the central visual element of the game. Each card displays the VTuber's portrait image from hololist.net, their name, agency, and variant tier badge. The card frame changes based on variant tier: Normal cards have a simple grey border, SR cards feature a silver border with a subtle glow effect, and SSR cards display a gold border with an animated shine effect. Cards are sized at approximately 120x160 pixels in the gallery grid and scale up to 200x280 pixels in pull results and detail views. The pull animation features a card flip reveal, where the card starts face-down and flips to reveal the character with a brief particle burst effect for SR and SSR pulls.

9.4 Responsive Design

The game is designed to be fully responsive, adapting seamlessly to screen sizes from 320px mobile screens to 1920px desktop displays. On mobile, the gallery grid displays 3 cards per row with a bottom tab bar for navigation. On tablet, the grid expands to 4-5 cards per row with a side tab bar. On desktop, the grid can display up to 8 cards per row with a horizontal top navigation bar. The Studio Room layout also adapts, showing stations in a vertical stack on mobile and a 2x2 grid on desktop. All interactive elements are sized with touch targets of at least 44x44 pixels to ensure comfortable mobile interaction.

10. Technical Architecture

10.1 Technology Stack

The game is built entirely with vanilla web technologies: HTML5 for structure, CSS3 for styling and animations, and JavaScript (ES6+) for game logic. No frameworks, libraries, or build tools are required, making the project accessible to developers of all skill levels and easy to deploy on any static hosting platform. The only external dependency is the character portrait images hosted on hololist.net, which are loaded dynamically via URLs stored in the game's character data file.

Technology	Purpose	Rationale
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HTML5	Page structure and semantic markup	Universal browser support, no build step
CSS3	Styling, animations, responsive layout	Card flip animations, dark theme, media queries
JavaScript ES6+	Game logic, state management, DOM manipulation	No framework overhead, full control
localStorage	Client-side persistence	No backend required, instant save/load
JSON	Character roster data format	Human-readable, easy to update roster
Firebase Realtime DB	Cloud save sync (free tier)	Cross-device save sync, no backend needed

Table 12: Technology Stack

10.2 File Structure

The project follows a clean, modular file structure that separates concerns for maintainability and readability. Each JavaScript file handles a distinct area of game functionality, and the character data is stored in a separate JSON file that can be updated independently of the game logic. This structure is designed to be approachable for developers learning web game development, with clear file names that indicate each file's purpose.

File	Purpose
index.html	Main game page, loads all scripts and stylesheets
css/style.css	All styles: dark theme, card frames, animations, responsive
js/data.js	Character roster data loader from JSON
js/game.js	Core game state, save/load, tick loop, currency calculations
js/gacha.js	Pull logic, rate calculation, pity system, duplicate handling
js/studio.js	Station assignment, upgrades, resource output calculation
js/characters.js	Levelling, ascension, shard conversion, variant management
js/ui.js	DOM rendering, tab navigation, pull animations, gallery grid
js/firebase.js	Firebase Realtime DB integration, cloud sync, save conflict resolution
data/characters.json	319 VTuber entries with name, image URL, agency, slug

Table 13: Project File Structure

10.3 Save/Load System

All game state is persisted to the browser's `localStorage` under a single JSON key. The save data includes the player's owned characters (with variant, level, and shard counts), currency balances, station assignments and upgrade levels, Studio Level and EXP, pity counter, daily login streak, last online timestamp (for offline earnings calculation), and profile statistics (total pulls, highest SSR streak, collection completion). The game auto-saves every 30 seconds and on every meaningful action (pull, level up, ascension, station change). A manual save button is also available. On load, the game calculates offline earnings based on the time elapsed since the last save and presents the accumulated resources to the player. In addition to local storage, the game supports cloud synchronisation via Firebase Realtime Database and manual export/import, as described in Section 10.5.

10.4 Game Tick System

A core game tick runs every second when the game is open, updating resource counters, calculating per-second income from all assigned stations, and updating the UI. When the game is closed, the last known timestamp is saved. Upon reopening, the game calculates the total offline earnings by multiplying the elapsed time (in seconds) by the per-second income rate and presents the result to the player as an "Offline Earnings" popup. Offline earnings are capped at a maximum of 12 hours to prevent excessive accumulation during extended absences, encouraging regular check-ins while still rewarding intermittent players.

10.5 Hybrid Save Sync System

The game employs a Hybrid Save Sync system (Option C) that combines automatic cloud synchronization via Firebase Realtime Database with manual export/import as a backup. This dual-layer approach ensures that players can seamlessly access their save data from any device while also having a reliable fallback if cloud services are unavailable. The design prioritises data safety, conflict resolution, and zero cost to both the developer and the players.

10.5.1 Architecture Overview

The save system operates on three layers. The first layer is `localStorage`, which provides instant local persistence and serves as the primary data source during gameplay. The second layer is Firebase Realtime Database, which automatically synchronises save data to the cloud using the player's unique Player ID as the database key. The third layer is the manual export/import system, which encodes the entire game state as a Base64 string that the player can copy and save externally. All three layers use the same save data format, ensuring full interoperability.

Layer	Method	Trigger	Purpose
1 (Primary)	localStorage	Auto-save every 30s + on every action	Instant local persistence
2 (Sync)	Firebase Realtime DB	Auto-push on save, auto-pull on load	Cross-device cloud sync
3 (Backup)	Export/Import (Base64)	Manual via Settings menu	Disaster recovery, device transfer

Table 16: Save System Architecture

10.5.2 Firebase Cloud Sync

Firebase Realtime Database (free Spark tier) is used for cloud synchronisation. The database is structured with each player's save data stored under a node keyed by their Player ID (format: MYVT-XXXX-XXXX-XXXX). When a player saves their game locally, the save data is automatically pushed to Firebase. When the game loads on a new device, it checks Firebase for existing save data under the entered Player ID and downloads it. The Firebase configuration is embedded directly in the client-side JavaScript, with database security rules set to allow read/write only for authenticated save data nodes.

The cloud sync flow operates as follows: on game load, if a Player ID is found in localStorage, the game checks Firebase for a cloud save. If the cloud save is newer than the local save (compared by lastOnline timestamp), the cloud version is downloaded and replaces the local data. If the local save is newer, the local data is pushed to Firebase. This timestamp-based conflict resolution ensures that the most recent save always wins, preventing accidental data loss when switching between devices.

10.5.3 Export/Import Backup

As a fallback and backup mechanism, players can manually export their save data as a Base64-encoded string from the Settings menu. This string contains the complete game state serialised as JSON and encoded to make it easy to copy and paste. Players are encouraged to save this string in a notes app, cloud storage, or message themselves the code. To import, the player pastes the code into the Import field, and the game validates and restores the full save state. This system serves as a critical backup for scenarios where Firebase is unavailable, the player loses access to their browser data, or they want to share their progress with a friend.

10.5.4 Cross-Device Transfer

To transfer progress to a new device, the player follows a simple two-step process. First, on the old device, they note their Player ID from the Home tab (or copy it to clipboard). Second, on the new device, they enter the Player ID in the Transfer Device

section of the Settings menu and click Download Save. The game then fetches the save data from Firebase using the provided Player ID, validates it, and restores the full game state. Alternatively, the player can use the manual export/import method if Firebase is not available. Both methods ensure that players never lose their progress when switching devices.

10.5.5 Cost and Scaling

The entire save sync system operates at zero cost. Firebase Realtime Database's free Spark tier provides 1 GB of stored data and 10 GB of monthly download bandwidth, which is more than sufficient for a browser game with moderate player counts. Each individual save state is approximately 5-50 KB of JSON data depending on collection size. Even with 1,000 active players saving every 30 seconds, the monthly bandwidth usage would remain well under the free tier limits. No server-side infrastructure, backend code, or database administration is required, keeping the project's \$0 budget commitment intact.

11. Character Data

The character roster is sourced from hololist.net, a comprehensive VTuber directory that lists 319 Malaysian VTubers as of the data extraction date. Each character entry includes their name, a unique slug identifier, a portrait image URL hosted on hololist.net, their agency affiliation, and a link to their hololist profile page. The data is stored in a JSON file (`characters.json`) that is loaded by the game at startup. This file can be easily updated to add new VTubers or modify existing entries without changing any game code.

11.1 Data Schema

Each character in the JSON file follows a consistent schema structure. The name field contains the VTuber's display name as listed on [hololist](https://hololist.net). The slug is a URL-friendly identifier derived from the name, used as a unique key for tracking ownership and progression. The image field contains the full URL to the VTuber's 300x300 portrait image on hololist.net. The agency field indicates their current affiliation, with "Independent" being the most common value (254 of 319 characters). The url field provides a direct link to the character's full hololist profile for players who want to learn more about a specific VTuber.

11.2 Agency Distribution

Agency	Count	Percentage
Independent	254	79.6%

VGakuenLive	7	2.2%
HoloDream	6	1.9%
Slice of Six Crew	5	1.6%
AmaiUsagi	4	1.3%
Others (37 agencies)	43	13.5%

Table 14: VTuber Agency Distribution (Top 5 + Others)

12. Sprint Development Plan

Development is organized into 8 sprints, each focusing on a self-contained feature area. Each sprint produces a playable, testable increment of the game. The sprints are ordered to build foundational systems first and progressively add complexity. After each sprint, a preview build will be provided for review and feedback. The estimated complexity ratings (Low, Medium, High) reflect the technical difficulty and scope of each sprint, helping to set expectations for review timelines.

Spr int	Feature	Deliverables	Complexity
1	Core UI Shell	Navigation tabs, dark theme, responsive layout, data loading	Low
2	Gacha Pull System	Banner UI, pull logic, rates, pity, card flip animation	Medium
3	Collection Gallery	Character grid, filters, detail popup, completion tracker	Medium
4	Studio Room	Station UI, character assignment, resource output display	Medium
5	Currency and Idle System	4 currencies, tick loop, offline earnings, daily login	Medium
6	Character Progression	Levelling, ascension, shard system, variant upgrades	High
7	Studio Progression	Studio levels, feature unlocks, station upgrades	Medium
8	Polish and Save/Load	Auto-save, manual save, stats, import/export, bug fixes	Low
9	Cloud Save Sync	Firebase Realtime DB integration, auto-sync, cross-device transfer	Medium

Table 15: Sprint Development Plan

12.1 Development Workflow

Each sprint follows a consistent workflow: (1) the AI agent implements the sprint's features based on this GDD, (2) a preview build is deployed for testing, (3) feedback is collected and addressed, and (4) the sprint is marked complete. The game owner reviews each preview build and provides feedback via simple commands such as "looks good" to approve or "change X" to request modifications. This lightweight review process ensures quality oversight without requiring deep technical involvement from the game owner, who can focus on design direction and player experience feedback.

13. Appendix: Key Data Structures

13.1 Game State Object

The following represents the top-level structure of the game state object that is serialized to localStorage. All nested objects follow the same pattern of storing character data keyed by their unique slug identifier.

```
{
  "version": 1,
  "currencies": { "stars": 1000, "starDust": 0, "starFragments": 0, "bondPoints": 0 },
  "characters": {
    "eri-james": { "owned": true, "variants": ["normal"], "level": 1, "shards": 0 },
    ...
  },
  "studio": {
    "level": 1, "exp": 0,
    "stations": {
      "streamRoom": { "level": 1, "assigned": null },
      ...
    }
  },
  "stats": { "totalPulls": 0, "ssrStreak": 0, "lastOnline": 1778900000 },
  "dailyLogin": { "streak": 0, "lastClaim": "2026-05-16" }
}
```